

AI Paper Roaster

Scientific Seminar Workflow

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Paper 1: CS-Guard — Benchmarking LLM Guardrails for Code Generation Security

Core Concept: The first benchmark specifically designed to test whether guardrails can prevent LLMs from generating malicious code. It shifts the guardrail conversation from natural-language moderation to technical cybersecurity and autonomous execution security.

Skimming Questions to Answer

- How is CS-Guard constructed? What data sources, attack categories, and evaluation metrics does it use?
- What exactly do the authors mean by “fictional software development scenarios,” and how do they bypass standard defenses?
- How do current guardrail frameworks perform on this benchmark — and where do they fail most catastrophically?

Paper 2: What the Guardrail Inspects Is Not What the Model Infers

Core Concept: Modern production guardrails are lightweight for latency reasons, so they have small context windows and chop long inputs into fragments. This creates an architectural blind spot: the guardrail approves fragments, while the downstream LLM reconstructs and executes a scattered attack.

Skimming Questions to Answer

- What is the exact mismatch between the guardrail's inspection window and the target LLM's context window?
- How do the authors scatter and hide adversarial instructions so each fragment looks benign in isolation?
- What is the attack success rate, and does it hold across different guardrail implementations and model families?

Round 1: Breakout Rooms & Initial Findings

- **Action:** Choose your assigned paper's breakout room.
- **Time:** 25 Minutes to read and answer the 3 skimming questions, mention a weak spot.
- **Deliverable:** Make notes in a .doc, .md, or draft a short presentation.

7-Minute Presentation

Elect **one speaker** to focus on high-impact insights.

Ask **1 tough question** to another team (but no answers yet!)

Round 2: Problem Solving & Pitch Creation

Time Allocated: 20 Minutes back in Breakout Rooms

Task A: Formulate Defense

Draft a rigorous answer to the cross-examination question thrown at your team by the audience during Round 1.

Task B: Build the Pitch Deck (3–5 Slides)

- Elaborate on a novel architectural idea or solution framework.
- Explicitly present the **weakness you exposed** in your paper.
- Demonstrate how your new design fixes or bypasses that exact loophole.

Round 3: The Defend & Pitch Showdown

Phase	Execution Rule
1. The Defense	Directly answer the question from the audience first.
2. The Pitch	You have exactly 3 minutes for your slide presentation.

Timer is Strict. Structure for Maximum Impact.